

SAT UZBEKISTAN

Telegram @satuzbhunters

SAT 23-August 2025

1.

Kaylani used fabric measuring 5 yards in length to make each suit for a men's choir. The relationship between the number of suits that Kaylani made, x , and the total length of fabric that she purchased y , in yards, is represented by the equation $y - 5x = 6$. What is the best interpretation of 6 in this context?

- (A) Kaylani made 6 suits.
- (B) Kaylani purchased a total of 6 yards of fabric.
- (C) Kaylani used a total of 6 yards of fabric to make the suits.
- (D) Kaylani purchased 6 yards more fabric than she used to make the suits.

2.

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 29.5% of the square footage of the farmland, and there are buildings on exactly 12.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?

3.

An entomologist placed an initial population of 40 *Tenebrio molitor*, a type of beetle, into a habitat and monitored the population over time. When the number of *Tenebrio molitor* in the habitat reached 160% of the initial population, the entomologist moved 75% of the *Tenebrio molitor* to a second habitat. How many *Tenebrio molitor* did the entomologist move to the second habitat at this time?

4.

x	y
0	n
4	$n + 11$
8	$n + 22$

There is a linear relationship between x and y . The table shows three values of x and their corresponding values of y in terms of a constant n . What is the slope of the line that represents this relationship in the xy -plane?

5.

$$5x + 9y = 20$$

$$rx + 18y = 40$$

In the given system of equations, r is a constant. The system has infinitely many solutions. What is the value of r ?

(A) 2

(B) 5

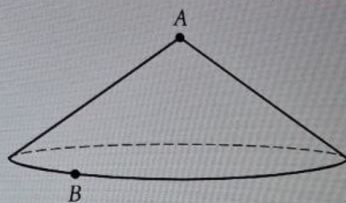
(C) 9

(D) 10

6.

For a certain type of rope, the equation $y = 900ax^2$, where a is a constant, gives the estimated breaking strength y , in pounds, of a rope with a circumference of x inches. Based on this equation, if a rope of this type has a circumference of 1.75 inches, it has an estimated breaking strength of 3,858.75 pounds. What is the estimated breaking strength, in pounds, of a rope of this type that has a circumference of 6.50 inches?

7.



Note: Figure not drawn to scale.

For the right circular cone shown, B is a point on the circumference of the base, and the length of segment AB (not shown) is 58 centimeters. If the height of the cone is 29 centimeters and the volume of the cone is $k\pi$ cubic centimeters, what is the value of k ?

8.

Line k is defined by $y = -\frac{23}{4}x + 1$. Line j is perpendicular to line k in the xy -plane. What is the slope of line j ?

9.

The binomial $x + 13y$ is a factor of which of the following?

(A) $x^2 + 14xy + 13y^2$

(B) $x^2 + 14xy + 14y^2$

(C) $x^2 + 39xy + 39y^2$

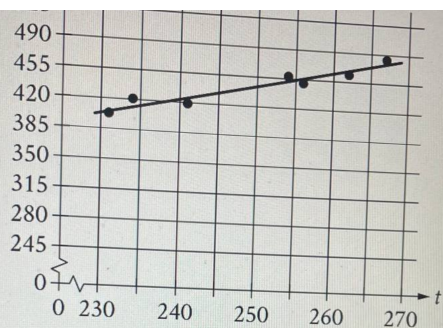
(D) $x^2 + 28xy + 42y^2$

10.

$$(x - k)^2 = (k - 5a)(x - k)$$

In the given equation, a and k are constants, where $k > 5a$. The sum of the solutions to the equation is $3k + 37$. What is the value of a ?

11.



Which of the following equations is the most appropriate linear model for the data shown?

(A) $d = -61.1 + 2.02t$

(B) $d = 161.1 + 2.02t$

(C) $d = 358.8 + 2.02t$

(D) $d = 393.8 + 2.02t$

12.

Two identical rectangular prisms each have a height of 40 centimeters (cm). The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{87}{47}K \text{ cm}^2$. What is the side length, in cm, of each square base?

(A) 4

(B) 7

(C) 14

(D) 28

13.

The table summarizes the distribution of age and assigned group for 105 participants in a study.

	0–9 years	10–19 years	20+ years	Total
Group A	5	19	11	35
Group B	4	8	23	35
Group C	26	8	1	35
Total	35	35	35	105

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

(A) $\frac{2}{7}$

(B) $\frac{3}{7}$

(C) $\frac{19}{35}$

(D) $\frac{6}{7}$

14.

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$150.00, zero products would be sold. For each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$2,187.50, which of the following could be the number of products sold? (revenue = price $\times$ number of products sold)

- (A) 25
- (B) 30
- (C) 815
- (D) 2,250

15.

18

Mark for Review

APC

The equation $d = 6t + 210$ gives the distance d , in feet, from a reference point on the surface of Mars to a Mars rover, where t is the number of minutes after the start of a period of observation. By how many feet does the Mars rover's distance from the reference point increase over a period of 20 minutes?

- (A) 330
- (B) 230
- (C) 120
- (D) 6

16.

During the first 5.00 seconds after a car started moving, its speed increased to 12.0 meters per second. From 5.00 seconds to 13.0 seconds after the car started moving, its speed increased from 12.0 meters per second to 25.0 meters per second. To the nearest hundredth, what is the positive difference between the average rate of change of the speed of the car, in meters per second per second, during the first 5.00 seconds after the car started moving and the average rate of change of the speed of the car, in meter per second per second, from 5.00 seconds to 13.0 seconds after the car started moving?

17.

An object is launched from a height of 121 feet above the ground. A quadratic function f models the height above the ground, in feet, of the object t seconds after it is launched. According to the model, 2 seconds after the object is launched, it reaches a maximum height of 185 feet above the ground. Which equation defines f ?

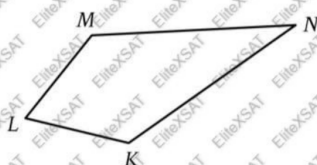
(A) $f(t) = -16(t + 185)^2 - 2$

(B) $f(t) = -16(t - 185)^2 + 2$

(C) $f(t) = -16(t + 2)^2 - 185$

(D) $f(t) = -16(t - 2)^2 + 185$

18.



Note: Figure not drawn to scale.

In quadrilateral $KLMN$ shown, $KL = 7$, $LM = 7$, $KN = 23$, and $MN = 23$. Diagonals KM and LN (not shown) intersect at point G (not shown), where $GK = 1$ and $GM = 1$. If the length of diagonal LN is $\sqrt{p} + \sqrt{w}$, where p and w are integers, what is the value of $p + w$?

19.

A circle in the xy -plane has its center at $(3, 7)$. Line t is tangent to this circle at the point $(a, -4)$, where a is a constant. The slope of line t is $\frac{5}{4}$. What is the value of a ?

A) $-\frac{43}{4}$

C) $\frac{59}{5}$

B) $-\frac{29}{5}$

D) $\frac{67}{4}$

20.

A circle in the xy -plane has its center at $(-2, -4)$. Line k is tangent to this circle at the point $(-5, -5)$. What is the slope of line k ?

A) -3

C) $\frac{1}{3}$

B) $-\frac{1}{3}$

D) 3

21.

A circle in the xy -plane has its center at $(-3, 8)$. Line k is tangent to this circle at the point $(-5, -9)$. What is the slope of line k ?

- A) -2 C) $\frac{1}{2}$
 B) $-\frac{1}{2}$ D) 2

22.

$$y = \frac{a-16}{x} + 4 \quad \text{2 unit down and 6 unit right} \quad \text{A) } y = \frac{(a-16)}{x-8}$$

B) $y = \frac{(a-16)}{x-6} + 2$

23.

$$f(x) = 5x^2 + 60x + 181$$

The function g is defined by $g(x) = f(x+8)$.

What is the minimum value of $g(x)$?

- A) -14 C) 1
 B) -6 D) 9

24.

Which of the following expressions has a factor of $x + 2b$, where b is a positive integer constant?

- A) $3x^2 + 7x + 14b$
 B) $3x^2 + 16x + 14b$
 C) $3x^2 + 18x + 14b$
 D) $3x^2 + 25x + 14b$

25.

A club plans to sell tote bags. The club members estimate they will sell 80 tote bags when the bags are priced at \$9 each. For every price increase of \$1, they estimate they will sell 8 fewer bags. What is the estimated revenue, in dollars, when the bags are priced at \$12 each? (revenue = price $\times$ number of bags sold)

26.

$$f(x) = a^x - b$$

The exponential function given above passes through points $(c, 12)$ and $(2c, 768)$. What is a possible value of b ?

27.

A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$150.00, zero products would be sold. For each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$2,187.50, which of the following could be the number of products sold? (revenue = price $\times$ number of products sold)

28.

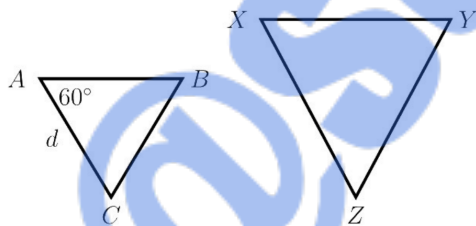
A company developed a plan to set the selling price of a product. The company determined that for a selling price of \$150.00, zero products would be sold. For each \$2.50 decrease in the selling price, the number of products sold would increase by one. For a revenue of exactly \$1,687.50, which of the following could be the number of products sold? (revenue = price $\times$ number of products sold)

29.

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 190)^\circ$. Which of the following could NOT be the sum of the measures of any two of these angles?

- A) $(-18x + 380)^\circ$
- B) $(-18x + 740)^\circ$
- C) $(18x - 380)^\circ$
- D) 180°

30.



Note: Figure not drawn to scale

For the triangles shown, triangle ABC dilated by a factor of 5 to obtain triangle XYZ , where $d = 19$. What is the measure, in degrees, of angle X ?

- A) 12
- B) 55
- C) 60
- D) 65

31.

On a plot of land, 52.0% of the square footage is farmland and the remaining square footage is pasture. There are buildings on exactly 21.5% of the square footage of the farmland, and there are buildings on exactly 14.0% of the square footage of the pasture. If there are buildings on exactly $p\%$ of the square footage of the plot of land, what is the value of p ?



32.

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k . Which of the following could **NOT** be equivalent to k , for all values of x ?

- A) $-14x + 1,540$
- B) $14x - 320$
- C) $-28x + 1,720$
- D) 360

33.

$x^2 + 84x + y^2 = 0$ find the centre of the circle.

- A) (-84;0) B) (42;0) C) (-42;0) D) (84;0)

34.

Four different research groups each surveyed a random sample of students from a certain high school to estimate the percentage of students at the school who speak a second language. The table shows the resulting estimates and their associated margins of error.

Group	Percentage of students who speak a second language	Margin of error
1	42%	13%
2	45%	12%
3	45%	8%
4	47%	15%

Assuming the margins of error were calculated in the same way, which group had the largest sample size?

(A) Group 1

(B) Group 2

(C) Group 3

(D) Group 4

C) Group 3

D) Group 4

35.

Javier deposits \$35 in a savings account at the end of each week. At the beginning of the last week of a year there was \$700 in that savings account. How much money, in dollars, will be in the account at the end of the 52th week of the year?

(A) 420

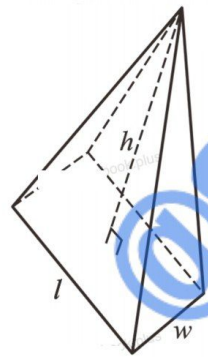
(B) 735

(C) 743

(D) 980

36.

36.



The figure shown is a right rectangular pyramid, where $l = 18$ units, $w = 9$ units, and $h = 12$ units. What is the surface area, in square units, of the pyramid?

pyramid?

37.

5 Mark for Review

Circles A and B are graphed in the xy -plane. Circle A is represented by the equation $(x - 10)^2 + (y + 10)^2 = 100$ and intersects the x -axis at the point (r, s) . Circle B has its center at (r, s) and has a radius of the same length as circle A . Which equation represents circle B ?

(A) $x^2 + (y - 10)^2 = 100$

(B) $x^2 + (y + 10)^2 = 100$

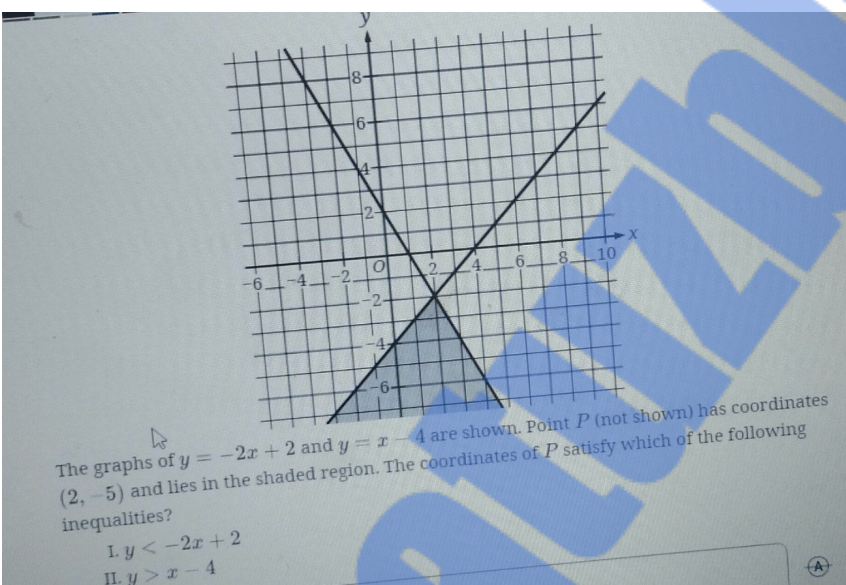
(C) $(x - 10)^2 + y^2 = 100$

(D) $(x + 10)^2 + y^2 = 100$

38.

A car dealership has only sedans, SUVs, and minivans for sale. On Monday, 20% of the vehicles for sale were sedans and 5% were SUVs. If there were 36 sedans for sale at the dealership on Monday, how many minivans were for sale?

39.



- A) I
- B) II
- C) I and II
- D) Neither I nor II

40. The table summarizes the distribution of age and assigned group for 90 participants in a study.

	0-9 years	10-19 years	20+ years	Total
Group A	15	11	4	30
Group B	4	5	21	30
Group C	11	14	5	30
Total	30	30	30	90

One of these participants will be selected at random. What is the probability of selecting a participant from group A, given that the participant is at least 10 years of age?

- A) $\frac{1}{6}$ B) $\frac{1}{4}$ C) $\frac{11}{30}$ D) $\frac{1}{2}$

41.

A circle in the xy -plane has its center at $(2,9)$. Line t is tangent to this circle at the point $(a,-4)$, where a is a constant. The slope of line t is $\frac{6}{5}$. what is the value of a ?

- A) $-\frac{68}{5}$ B) $-\frac{53}{6}$ C) $\frac{77}{6}$ D) $\frac{88}{5}$

42.

A team housing an event to raise money for new uniforms plans to sell at least 140 tickets before this event and at least 220 tickets during this event to raise a total of at least \$5,820 from all tickets sold. The price of a ticket during this event is \$3 less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

- A) $140(x+3)+220x \leq 5,820$ C) $140(x+3)+220x \geq 5,820$
B) $140(x-3)+220x \leq 5,820$ D) $140(x-3)+220x \geq 5,820$

43.

In the given expression, b is a positive integer if $qz^6 + r$ is a factor of the expression, where q and r are positive integers, what is the greatest possible value of b ?

$$26z^{12} + bz^6 + 70$$



44.

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x-520)$. the sum of the measures of one of the acute angles and three obtuse angles is $(18x+w)$. What is the value of w ?



@saturchunters